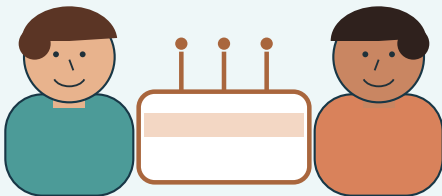


START HERE

How many people does it take before there's a 50/50 chance at least two share a birthday?



12

23

56

182

253

COMMIT

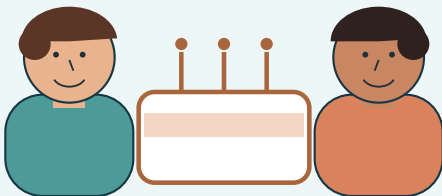
Commit to one answer before Card 2.

FIRST REVEAL

23

people

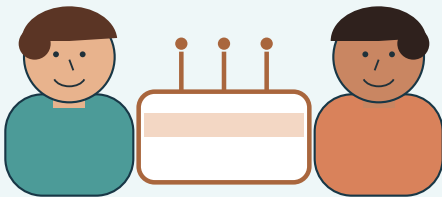
With 23 people, the chance is already
about 50.7%.



How many people does it take before
there's more than a 99% chance?

COMMIT AGAIN

How many people does it take before there's more than a 99% chance at least two share a birthday?



31

57

104

253

350

COMMIT

Commit to one answer before Card 4.

SECOND REVEAL

57

people

**With only 57 people, the chance is already
just over 99%.**

**How can the probability grow
that quickly?**

What might be hiding inside the group?

COMMIT

Commit to an explanation before Card 5.

WHAT WAS HIDING?

253

different pairs among 23 people

$$22 + 21 + 20 + \dots + 2 + 1$$
$$= 253$$

Each pair is another opportunity for two birthdays to

253 opportunities for a match.

QUICK TRANSFER

Would the answer still be
23 for wedding
anniversaries?



YES

NO

COMMIT

Commit before Card 7.

ANNIVERSARY REVEAL

2**can already be enough**

If a married couple is in the group, those two already share a wedding anniversary.

64 teams enter a single-elimination tournament.

How many games must be played to crown one champion?

COMMIT

Commit to a number before Card 8.

TOURNAMENT REVEAL

64 -> 32 -> 16 -> 8 -> 4 -> 2 -> 1

Each round cuts the field in half.

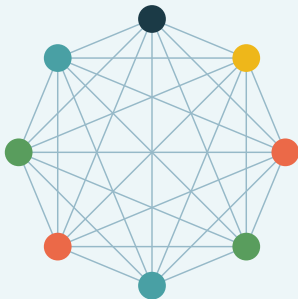
63

games

Every game eliminates one team, so 63 teams must lose to leave one champion.

SAME 64 TEAMS. DIFFERENT RULE.

Now every team plays every other team exactly once.



A few teams shown - every pair connected

How many games will be played?

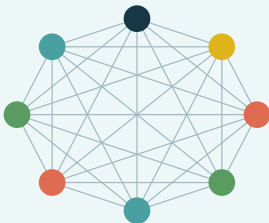
COMMIT

Estimate. Commit before Card 10.

SECOND TOURNAMENT REVEAL

2,016

games



Same 64 teams. Why did the number of games jump from 63 to 2,016?

COMMIT

Explain the difference before Card 11.

LOOK BACK

23 PEOPLE

253 possible pairs

64 TEAMS

2,016 possible pairings

The number of people or teams wasn't the surprising part.

What was? Where else might something important be hiding in relationships between things?

BEHIND THE EXPERIENCE · READ LAST

WHY THE NUMBERS GROW

1. PAIRS GROW FASTER THAN GROUPS

With 23 people, there are 253 different pairs. With 64 teams, there are 2,016 different pairings.

$$23 \times 22 / 2 = 253$$

$$64 \times 63 / 2 = 2,016$$

2. WHY DIVIDE BY 2?

A-B and B-A are the same pair. Multiplying counts each pair twice, so divide by 2.

3. WHY BIRTHDAYS RISE SO QUICKLY

Each new pair is another opportunity for a birthday match. The exact 50.7% comes from first finding the chance that all 23 birthdays are different, then subtracting that from 100%.